Lakshmi Ghosh

Embedded Firmware Engineer

RESUME_1108

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Total Experience: 12 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 12 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: C++, JTAG, FreeRTOS, Firmware Debugging, C, ARM Cortex-M, Git, UART, CAN, I2C, SPI, STM32

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | Ridgeview Engineering | Jan 2022 - Present

- Lowered field-reset incidents by 22% with watchdog recovery and diagnostics
- Improved battery runtime by 39% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 35% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | Meridian Vector Labs | Jan 2018 - Dec 2021

- Reduced embedded task jitter by 40% using interrupt priority and queue tuning
- Validated 608 hardware test vectors across protocol and timing corner cases

Associate Embedded Firmware Engineer | NovaPeak Engineering | Jul 2014 - Dec 2017

- Lowered field-reset incidents by 27% with watchdog recovery and diagnostics
- Improved battery runtime by 22% with peripheral and sleep-state optimization

PROJECTS

Industrial Sensor Firmware | CAN, STM32, JTAG, SPI

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | Firmware Debugging, I2C, STM32, FreeRTOS

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Pragati Institute of Applied Sciences, 2014
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2026)